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Statistics for Business Administration

Exam: MA9712 / Practice Exam B – Solutions **Date:** Wednesday 5th August, 2026
Examiner: Prof. Mathias Drton **Time:** 14:00 – 15:30

Working instructions

- This exam consists of pages with a total of 9 problems.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is **52 credits**.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one scientific pocket calculator
 - the cheat sheet, which is contained in this exam on page 16
- Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____

Early submission at _____

Problem 1

7.5 credits

A university surveys $n = 600$ students about their satisfaction with campus dining (*Satisfied*, *Neutral*, *Dissatisfied*) and records each student's meal-plan type (*Full*, *Partial*, *None*). The observed counts are:

Meal plan	Satisfied	Neutral	Dissatisfied	Total
Full	150	60	30	240
Partial	110	70	30	210
None	40	50	60	150
Total	300	180	120	600

- a) Identify the response variable and the explanatory variable, and state the type (categorical/numerical) of each.

Solution. Response variable: *satisfaction* (categorical, ordinal). Explanatory variable: *meal-plan type* (categorical, nominal). We study how satisfaction depends on meal plan.

- b) The students were selected by a simple random sample of the university's student body, but were not randomly assigned to a meal-plan type. Can the results be generalized to the whole student body? Can a causal claim "meal plan affects satisfaction" be made? Justify both answers.

Solution. Yes to generalization: because the sample is a random sample of the student body, the results generalize to that population. No to causation: this is an observational study (no random assignment to meal plans), so a causal claim cannot be made — confounding variables (e.g. budget, on/off-campus living) may drive both meal-plan choice and satisfaction.

- c) Which single graphical display is most appropriate to compare the satisfaction distribution across the three meal-plan groups? Name it.

Solution. A **segmented (stacked) bar chart of conditional proportions** (side-by-side/segmented bar chart), one bar per meal-plan group showing the satisfaction breakdown. (A mosaic plot is equally acceptable.)

- d) Compute the joint probability that a randomly selected student has *no* meal plan and is *Dissatisfied*.

Solution. $\mathbb{P}(\text{None} \cap \text{Dis}) = \frac{60}{600} = 0.100$. $\mathbb{P}(\text{None} \cap \text{Dis}) = 0.100$

- e) Show that the probability that a randomly selected student is *Satisfied* or has a *Full* meal plan equals 0.650 (to three decimal places).

Solution. $\mathbb{P}(\text{Sat} \cup \text{Full}) = \mathbb{P}(\text{Sat}) + \mathbb{P}(\text{Full}) - \mathbb{P}(\text{Sat} \cap \text{Full}) = \frac{300}{600} + \frac{240}{600} - \frac{150}{600} = \frac{390}{600} = 0.650$.

- f) Suppose 8 students are drawn independently from the population, in which the proportion of *Dissatisfied* students is $p = 0.20$. Let X be the number of *Dissatisfied* students among the 8. Compute $\mathbb{P}(X = 3)$. You may use the hints:

```
dbinom(3, size = 8, prob = 0.2) ## [1] 0.1468
pbinom(3, size = 8, prob = 0.2) ## [1] 0.9437
pbinom(2, size = 8, prob = 0.2) ## [1] 0.7969
```

Solution. $X \sim \text{Bin}(8, 0.2)$, so $\mathbb{P}(X = 3) = \binom{8}{3}(0.2)^3(0.8)^5 = 56 \cdot 0.008 \cdot 0.32768 = 0.1468$. (Equivalently $\text{pbinom}(3) - \text{pbinom}(2) = 0.9437 - 0.7969 = 0.1468$.) $\mathbb{P}(X = 3) \approx 0.147$

Problem 2

6 credits

A consumer group measures the battery life (in hours, under a standardized test) of $n = 50$ smartphones of one model. The sample has mean $\bar{x} = 18.0$ hours and standard deviation $s = 4.0$ hours. A boxplot of the data shows the first quartile at $Q_1 = 15.5$ hours and the third quartile at $Q_3 = 21.0$ hours.

- a) Read the interquartile range (IQR) of battery life from the boxplot information.

Solution. $\text{IQR} = Q_3 - Q_1 = 21.0 - 15.5 = 5.5$ hours.

- b) State the conditions required by the Central Limit Theorem so that $\bar{x}$ is approximately normally distributed, and argue whether they are satisfied here.

Solution. (i) *Independence*: observations independent — satisfied if the phones are a random sample and $n = 50 < 10\%$ of all phones of this model. (ii) *Sample size / skew*: $n = 50 \geq 30$ is large enough for the CLT to apply even if battery life is moderately skewed. Both conditions are reasonably met, so $\bar{x}$ is approximately normal.

- c) A 90% confidence interval for the mean battery life μ is reported as $[17.07, 18.93]$ hours. Justify this interval by computing the standard error and margin of error. You may use $z_{0.95} = 1.645$.

Solution. $\text{SE}(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{4.0}{\sqrt{50}} = 0.5657$. Margin of error $= z^* \text{SE} = 1.645 \cdot 0.5657 = 0.931$. $\text{CI} = \bar{x} \pm \text{ME} = 18.0 \pm 0.931 = [17.07, 18.93]$. $[17.07, 18.93]$

- d) Interpret the confidence interval in context in one sentence.

Solution. We are 90% confident that the true mean battery life of this smartphone model is between 17.07 and 18.93 hours.

Problem 3

4 credits

A rapid test for a medical condition is evaluated. In the target population the prevalence of the condition is 4%. The test has sensitivity 92% (probability of a positive result given the condition is present) and specificity 95% (probability of a negative result given the condition is absent).

- a) Define suitable events for “has the condition” and “tests positive”, and describe the probability tree (branches and their probabilities).

Solution. Let $D = \{\text{has condition}\}$, $D^c = \{\text{no condition}\}$, $T = \{\text{tests positive}\}$. First split: $\mathbb{P}(D) = 0.04$, $\mathbb{P}(D^c) = 0.96$. Second split (sensitivity/specificity): $\mathbb{P}(T | D) = 0.92$, $\mathbb{P}(T^c | D) = 0.08$; $\mathbb{P}(T | D^c) = 1 - 0.95 = 0.05$, $\mathbb{P}(T^c | D^c) = 0.95$.

- b) Compute the overall probability that a randomly tested person receives a positive test result.

Solution. Law of total probability: $\mathbb{P}(T) = \mathbb{P}(T | D)\mathbb{P}(D) + \mathbb{P}(T | D^c)\mathbb{P}(D^c) = 0.92 \cdot 0.04 + 0.05 \cdot 0.96 = 0.0368 + 0.0480 = 0.0848$.

- c) Compute the positive predictive value, i.e. the probability that a person actually has the condition given that they tested positive. Round to three decimal places.

Solution. $\mathbb{P}(D | T) = \frac{\mathbb{P}(T | D)\mathbb{P}(D)}{\mathbb{P}(T)} = \frac{0.0368}{0.0848} = 0.4340 \approx 0.434$. $\mathbb{P}(D | T) \approx 0.434$

Problem 4

5 credits

A clinical trial randomly assigns 400 volunteers to receive either a new vaccine ($n_1 = 200$) or a placebo ($n_2 = 200$). A mild side effect is reported by 46 people in the vaccine group and by 28 people in the placebo group. Researchers ask whether the side-effect rate differs between the two treatments.

- a) Let p_1 and p_2 denote the true side-effect proportions in the vaccine and placebo populations. State the null and alternative hypotheses for a two-sided test.

Solution. $H_0 : p_1 = p_2$ vs. $H_A : p_1 \neq p_2$.

- b) Compute the observed difference in sample proportions $\hat{p}_1 - \hat{p}_2$.

Solution. $\hat{p}_1 = \frac{46}{200} = 0.23$, $\hat{p}_2 = \frac{28}{200} = 0.14$, $\hat{p}_1 - \hat{p}_2 = 0.23 - 0.14 = 0.09$.

- c) To obtain a p -value, the treatment labels were randomly reshuffled 10,000 times and the difference recomputed each time, giving the null distribution summarized below. Report the p -value.

```
# number of reshuffles with |difference| >= observed difference
sum(abs(perm_diffs) >= abs(obs_diff)) ## [1] 214
length(perm_diffs)                    ## [1] 10000
```

Solution. $p\text{-value} = \frac{214}{10000} = 0.0214$.

- d) State your conclusion at significance level $\alpha = 0.05$ in context.

Solution. Since the p -value 0.0214 is less than the significance level 0.05, we reject H_0 . The data provide convincing evidence that the side-effect rate differs between the vaccine and the placebo.

Problem 5

4 credits

For each of the following statements, decide whether it is **true** or **false** and *justify your answer*. Only correct justifications are awarded with points, not the decision (true/false).

- a) A p -value of 0.03 means that the probability the null hypothesis is true equals 0.03.

Solution. False. The p -value is the probability of observing data at least as extreme as the observed data *assuming* H_0 is true; it is not $\mathbb{P}(H_0 \text{ true})$. H_0 is either true or false — it is not a random event with a probability in the frequentist framework.

- b) If we increase the confidence level from 90% to 99% while keeping the same data, the confidence interval becomes wider.

Solution. True. A higher confidence level uses a larger critical value z^* ($1.645 \rightarrow 2.576$), which enlarges the margin of error $z^* \text{SE}$ and widens the interval.

- c) For a fixed confidence level, quadrupling the sample size halves the margin of error of a confidence interval for a mean.

Solution. True. $\text{ME} = z^* s / \sqrt{n} \propto 1 / \sqrt{n}$. Replacing n by $4n$ multiplies the ME by $1 / \sqrt{4} = 1/2$, i.e. halves it.

- d) The success–failure condition $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$ is used to justify the normal approximation for inference about a proportion.

Solution. True. The condition ensures enough expected successes and failures for the sampling distribution of $\hat{p}$ (a binomial count divided by n) to be well approximated by a normal distribution.

Problem 6

4 credits

A genetics experiment crosses two pea-plant lines. Mendelian theory predicts that the four phenotype classes (call them A, B, C, D) occur in the ratio 9 : 3 : 3 : 1. Among $n = 160$ offspring the observed counts are:

Phenotype	A	B	C	D	Total
Observed	100	26	28	6	160

Test at $\alpha = 0.05$ whether the data are consistent with the predicted 9 : 3 : 3 : 1 ratio.

- a) State the null and alternative hypotheses for the goodness-of-fit test.

Solution. H_0 : the phenotype probabilities are $(\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16})$, i.e. the 9 : 3 : 3 : 1 ratio holds.
 H_A : at least one phenotype probability differs from this ratio.

- b) Compute the expected counts under H_0 and verify the condition that every expected count is at least 5.

Solution. $E_A = 160 \cdot \frac{9}{16} = 90$, $E_B = 160 \cdot \frac{3}{16} = 30$, $E_C = 30$, $E_D = 160 \cdot \frac{1}{16} = 10$. All expected counts (90, 30, 30, 10) are ≥ 5 , so the chi-square approximation is valid.

- c) Compute the chi-square test statistic, state the degrees of freedom, and bracket the p -value using the hints below. Conclude in context.

```
qchisq(0.90, df = 3) ## [1] 6.251
qchisq(0.95, df = 3) ## [1] 7.815
qchisq(0.99, df = 3) ## [1] 11.345
```

Solution. $\chi^2 = \sum \frac{(O - E)^2}{E} = \frac{(100 - 90)^2}{90} + \frac{(26 - 30)^2}{30} + \frac{(28 - 30)^2}{30} + \frac{(6 - 10)^2}{10} = \frac{100}{90} + \frac{16}{30} + \frac{4}{30} + \frac{16}{10} = 1.111 + 0.533 + 0.133 + 1.600 = 3.378$. Degrees of freedom $df = k - 1 = 4 - 1 = 3$. Since $\chi^2 = 3.378 < 6.251 = \text{qchisq}(0.90, 3)$, the p -value > 0.10 . Because the p -value is larger than 0.05, we fail to reject H_0 . The data do not provide convincing evidence that the phenotype ratio differs from 9 : 3 : 3 : 1. $\chi^2 = 3.378$, $df = 3$, $p > 0.10$

Problem 7

9 credits

An education researcher compares final-exam scores (points, out of 100) of students taught by three methods: *Lecture*, *Flipped*, and *Online*. Each group has $n_i = 7$ students ($N = 21$). The group summary is:

Method	n_i	mean $\bar{y}_i$	sd s_i
Lecture	7	72	$\sqrt{20}$
Flipped	7	81	$\sqrt{20}$
Online	7	75	$\sqrt{20}$

The grand mean is $\bar{y} = 76$.

- a) State the null and alternative hypotheses of the one-way ANOVA.

Solution. $H_0 : \mu_{\text{Lec}} = \mu_{\text{Flip}} = \mu_{\text{Onl}}$ vs. $H_A : \text{at least one group mean differs from the others.}$

- b) State the three conditions required for ANOVA and comment briefly on each.

Solution. (i) *Independence* within and between groups — plausible if students were randomly assigned to methods. (ii) *Approximate normality* of scores within each group — reasonable for exam scores, important at small $n_i = 7$. (iii) *Constant variance* across groups — satisfied here since all groups have the same sample sd $s_i = \sqrt{20}$.

- c) Fill in the missing (NA) entries of the ANOVA table below.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
method	2	NA	NA	NA	NA
Residuals	18	360	NA		

Solution. $\text{SSG} = \sum n_i(\bar{y}_i - \bar{y})^2 = 7[(72 - 76)^2 + (81 - 76)^2 + (75 - 76)^2] = 7(16 + 25 + 1) = 7 \cdot 42 = 294$. $\text{MSG} = \frac{\text{SSG}}{k - 1} = \frac{294}{2} = 147$. $\text{MSE} = \frac{\text{SSE}}{N - k} = \frac{360}{18} = 20$. $F = \frac{\text{MSG}}{\text{MSE}} = \frac{147}{20} = 7.35$.
Completed table:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
method	2	294	147	7.35	0.00464
Residuals	18	360	20		

- d) Carry out the F -test at $\alpha = 0.05$ and state your conclusion. You may use:

```
qf(0.95, df1 = 2, df2 = 18) ## [1] 3.555
qf(0.99, df1 = 2, df2 = 18) ## [1] 6.013
```

Solution. $F = 7.35$ with $df_1 = 2$, $df_2 = 18$. Since $F = 7.35 > 6.013 = \text{qf}(0.99, 2, 18)$, the p -value $< 0.01 < 0.05$, so we reject H_0 . The data provide convincing evidence that at least one teaching method has a different mean exam score.

- e) Compute the proportion of variability in exam scores explained by teaching method ($R^2 = \text{SSG}/\text{SST}$).

Solution. $\text{SST} = \text{SSG} + \text{SSE} = 294 + 360 = 654$. $R^2 = \frac{\text{SSG}}{\text{SST}} = \frac{294}{654} = 0.4495 \approx 44.95\%$.

$$R^2 \approx 0.4495$$

- f) The three pairwise comparisons are carried out with Bonferroni-adjusted p -values shown below. State the Bonferroni-adjusted significance level and which pairs of methods differ significantly at family-wise level 0.05.

	diff	p.adj
Flipped-Lecture	9	0.00425
Flipped-Online	6	0.06555
Online-Lecture	3	0.67660

Solution. With $K = \binom{3}{2} = 3$ comparisons, the Bonferroni-adjusted per-test level is $\alpha^* = 0.05/3 = 0.0167$ (equivalently, compare the reported adjusted p -values directly to 0.05). Only *Flipped vs. Lecture* ($p_{\text{adj}} = 0.00425 < 0.05$) differs significantly; *Flipped-Online* (0.066) and *Online-Lecture* (0.677) do not.

Problem 8

8.5 credits

A company models monthly salary (in thousands of EUR) of $n = 120$ employees using years of **experience** and **gender** (*Female* is the baseline; **genderMale** = 1 for male employees), *including an interaction term*. The fitted `lm()` summary is shown below.

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      32.00      1.20      ?    < 2e-16
experience         1.50      0.18      ?    < 2e-16
genderMale        3.00      1.60      ?         ?
experience:genderMale 0.80      0.30      ?         ?
---
Residual standard error on 116 degrees of freedom

```

- a) Write the fitted regression equation for a general employee (using the indicator **genderMale**).

Solution. $\widehat{\text{salary}} = 32.00 + 1.50 \cdot \text{experience} + 3.00 \cdot \text{genderMale} + 0.80 \cdot (\text{experience} \times \text{genderMale})$.

- b) Simplify the fitted equation for *male* employees (**genderMale**= 1).

Solution. Set **genderMale** = 1: $\widehat{\text{salary}} = (32.00 + 3.00) + (1.50 + 0.80) \cdot \text{experience} = 35.00 + 2.30 \cdot \text{experience}$.
 $\widehat{\text{salary}}_{\text{male}} = 35.00 + 2.30 \cdot \text{experience}$

- c) Interpret the **experience** coefficient (1.50) for female employees in context.

Solution. All else equal, for female employees each additional year of experience is associated with an increase of 1.50 thousand EUR (1500 EUR) in predicted monthly salary.

- d) Interpret the interaction coefficient (0.80): how does the effect of one additional year of experience differ between male and female employees?

Solution. The per-year experience effect for males is $1.50 + 0.80 = 2.30$ thousand EUR, i.e. 0.80 thousand EUR (800 EUR) *more* per year than for females. The return to experience is steeper for male employees.

- e) Test at $\alpha = 0.05$ whether the interaction term is significant. Compute its t -value and use the hint below to obtain the p -value; then conclude.

```
| 2 * pt(2.667, df = 116, lower.tail = FALSE) ## [1] 0.00876
```

Solution. $H_0 : \beta_{\text{int}} = 0$ vs. $H_A : \beta_{\text{int}} \neq 0$. $t = \frac{b_{\text{int}}}{\text{SE}} = \frac{0.80}{0.30} = 2.667$ with $df = n - k - 1 = 120 - 3 - 1 = 116$. From the hint the two-sided p -value is 0.00876. Since $0.00876 < 0.05$, we reject H_0 : the interaction between experience and gender is statistically significant.

Problem 9

5 credits

A telecom company models customer **churn** (*yes/no*) using **tenure** (months), **monthly_charges** (EUR), and **contract_type** (baseline *Month-to-month*; levels *One year*, *Two year*). The fitted logistic-regression model is shown below.

```

glm(formula = churn ~ tenure + monthly_charges + contract_type,
     family = _____, data = customers)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)

```

(Intercept)	-0.5200	0.2100	-2.48	0.0131
tenure	-0.0450	0.0060	-7.50	< 2e-16
monthly_charges	0.0310	0.0040	7.75	< 2e-16
contract_typeOne year	-0.9100	0.1500	-6.07	< 2e-16
contract_typeTwo year	-1.7600	0.2200	-8.00	< 2e-16

- a) Fill in the blank in the `glm()` call so that a logistic regression is fitted.

Solution. `family = binomial` (equivalently `binomial(link = "logit")`).

- b) Interpret the `tenure` coefficient (-0.0450) as an odds ratio, in context.

Solution. $OR = e^{-0.0450} = 0.956$. All else equal, each additional month of tenure multiplies the odds of churn by 0.956, i.e. the odds of churn is decreased by about 4.4% per extra month.

$$OR = e^{-0.045} \approx 0.956$$

- c) Using a 0.5 probability cut-off, the model produces the confusion matrix below (rows = prediction, columns = actual). Compute the sensitivity and the specificity.

	Actual: No	Actual: Yes
Pred: No	820	90
Pred: Yes	80	210

Solution. Treat churn = “Yes” as positive. $TP = 210$, $FN = 90$, $TN = 820$, $FP = 80$. Sensitivity $= \frac{TP}{TP + FN} = \frac{210}{300} = 0.700$. Specificity $= \frac{TN}{TN + FP} = \frac{820}{900} = 0.911$. sens = 0.700, spec = 0.911

- d) The full model has an ROC area under the curve (AUC) of 0.861; a reduced model that drops `contract_type` has AUC 0.833. Which model discriminates churners from non-churners better, and why?

Solution. The full model, because its AUC (0.861) is larger than the reduced model’s (0.833). A higher AUC means a higher probability that a randomly chosen churner receives a larger predicted churn probability than a randomly chosen non-churner, i.e. better discrimination.

Cheat sheet

Exploring data. $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$, $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$, $s = \sqrt{s^2}$, IQR = $Q_3 - Q_1$, fences $Q_1 - 1.5 \text{ IQR}$, $Q_3 + 1.5 \text{ IQR}$.

$$r_{(x,y),n} = \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{s_x s_y}$$

Probability. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$; $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ if independent; $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$;
 $\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$.

$$\mathbb{E}(X) = \sum_i x_i \mathbb{P}(X = x_i) \text{ (discrete)}, \quad \mathbb{E}(X) = \int x f(x) dx \text{ (cont.)},$$

$$\text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2, \quad \text{SD}(X) = \sqrt{\text{Var}(X)}, \quad \text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y), \quad Z = \frac{X - \mathbb{E}(X)}{\sqrt{\text{Var}(X)}}.$$

Distributions.

$$\text{Geometric: } \mathbb{P}(X = x) = (1-p)^{x-1}p, \quad \mathbb{E}(X) = \frac{1}{p}, \quad \text{SD}(X) = \sqrt{\frac{1-p}{p^2}}.$$

$$\text{Binomial: } \mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad \mathbb{E}(X) = np, \quad \text{SD}(X) = \sqrt{np(1-p)}.$$

$$\text{Neg. Binomial: } \mathbb{P}(X = x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}, \quad x \geq k. \quad \text{Poisson: } \mathbb{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad \mathbb{E}(X) = \lambda, \quad \text{SD}(X) = \sqrt{\lambda}.$$

Inference — categorical.

$$1 \text{ prop. CI: } \hat{p} \pm z^* \text{SE}(\hat{p}), \quad \text{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}; \quad \text{test: } Z = \frac{\hat{p} - p_0}{\text{SE}_0}, \quad \text{SE}_0 = \sqrt{\frac{p_0(1-p_0)}{n}}.$$

$$2 \text{ prop. CI: } (\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}; \quad \text{test: } \text{SE}_0 = \sqrt{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}, \quad \hat{p}_{\text{pool}} = \frac{\hat{p}_1 n_1 + \hat{p}_2 n_2}{n_1 + n_2}.$$

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}, \quad \text{expected}_{ij} = \frac{(\text{row } i)(\text{col } j)}{\text{total}}, \quad df = k - 1 \text{ (GOF) or } (r - 1)(c - 1).$$

Inference — numerical.

$$1\text{-sample } t: \bar{x} \pm t_{df}^* \text{SE}(\bar{x}), \quad T_{df} = \frac{\bar{x} - \mu_0}{\text{SE}(\bar{x})}, \quad \text{SE}(\bar{x}) = \frac{s}{\sqrt{n}}, \quad df = n - 1.$$

$$2\text{-sample } t: (\bar{x}_1 - \bar{x}_2) \pm t_{df}^* \text{SE}, \quad T_{df} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\text{SE}}, \quad \text{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \quad df = \min(n_1 - 1, n_2 - 1).$$

$$\text{Sample size: two-sample } n \geq \frac{2\sigma_0^2}{\delta_0^2} (z_{1-\alpha^*} + z_{1-\beta})^2; \quad \text{one-sample } n \geq \frac{\sigma_0^2}{\delta_0^2} (z_{1-\alpha^*} + z_{1-\beta})^2, \quad \alpha^* \in \{\alpha, \alpha/2\}.$$

$$\text{ANOVA \& regression. } F = \frac{\text{MSG}}{\text{MSE}}, \quad df_1 = k - 1, \quad df_2 = n - k.$$

$$\text{SSG} = \sum_{i=1}^k n_i (\bar{y}_i - \bar{y})^2, \quad \text{SST} = \sum_{j=1}^n (y_j - \bar{y})^2, \quad \text{SSE} = \text{SST} - \text{SSG}.$$

$$\text{Post-hoc } T_{df} = \frac{\bar{x}_i - \bar{x}_j}{\sqrt{\text{MSE}/n_i + \text{MSE}/n_j}}, \quad df = n - k; \quad \text{Bonferroni } \alpha^* = \alpha/K, \quad p_{\text{adj},i} = K p_i, \quad K = \binom{k}{2}.$$

$$\hat{y} = b_0 + b_1 x, \quad b_1 = r_{(x,y),n} \frac{s_y}{s_x}, \quad \hat{\sigma} = \sqrt{\frac{1}{n-2} \sum (y_i - \hat{y}_i)^2}, \quad T_{df} = \frac{b_1 - \beta_{1,0}}{\text{SE}_{b_1}}, \quad df = n - 2.$$

$$R^2 = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}, \quad R_{\text{adj}}^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2 / (n - k - 1)}{\sum (y_i - \bar{y})^2 / (n - 1)}, \quad \text{AIC} = c + n \ln(\sum e_i^2) + 2k.$$

Logistic regression. $\text{logit}(p_i) = \log \frac{p_i}{1-p_i} = \beta_0 + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i}$, $\text{OR}_j = e^{\beta_j}$, sensitivity = $\mathbb{P}(\text{Test+} | \text{Cond+})$, specificity = $\mathbb{P}(\text{Test-} | \text{Cond-})$.

Quantile hints (standard normal). $z_{0.70} = 0.524$, $z_{0.80} = 0.842$, $z_{0.90} = 1.282$, $z_{0.95} = 1.645$, $z_{0.975} = 1.960$, $z_{0.99} = 2.326$, $z_{0.995} = 2.576$.